

Q: "Which of these objects (heater, trash can, door, cup) is the closest to the microwave?"

(a) Frame prompting: every candidate object sits in its own frame — no pairwise distance is ever visible



camera
motion?
----->



camera
motion?
----->

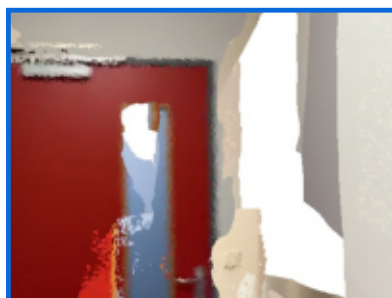


the microwave, door, heater (and cup, trash can) each appear at a different depth in a different frame; comparing their distances requires fusing all views into one 3-D layout — implicitly
—**unreliable for (small) VLMs**

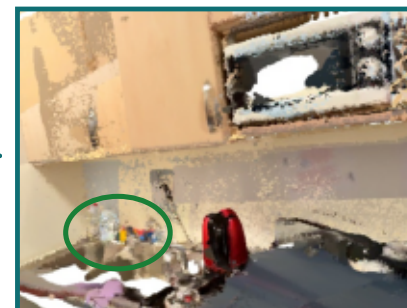
(b) ViewTree-D: walk to the reference object; the nearest candidate becomes directly visible



VGGT
(frozen)



walk (beam over actions):
start at spot 3 →
TURN RIGHT →
FORWARD to spot 8



answer: D (cup) ✓
consensus of the retained depth-3
walks, confidence above direct